Blessed G/Michael Catholic Secondary School

Semester 1 Physics Mid1 Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade11 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** In two-dimensional projectile motion, which of the following quantities remains constant throughout the flight (neglecting air resistance)?

A. Vertical velocity **B.** Horizontal acceleration

C. Horizontal velocity **D.** Total momentum direction

____ **2.** Given vector A = 3i + 4j and vector B = 2i - j, what is the dot product A · B?

A. 2 **B.** 6 **C.** 10 **D.** 14

____ **3.** A projectile is launched with velocity 50 m/s at an angle of 30 degrees above horizontal. What is its initial vertical velocity component vy? (sin 30° = 0.5)

A. 25 m/s **B.** 43.3 m/s **C.** 50 m/s **D.** 10 m/s

____ **4.** In uniform circular motion with radius r and constant speed v, the centripetal acceleration is directed:

A. Radially outward from the center
B. Radially inward toward the center
C. Tangential to the circle in the direction of motion
D. Opposite to the velocity vector

____ **5.** A 2 kg mass attached to a string moves in a circle of radius 0.5 m with a speed of 4 m/s. What is the centripetal force?

A. 16 N **B.** 32 N **C.** 64 N **D.** 8 N

____ **6.** Work done by a conservative force along any closed path is always:

A. Maximum **B.** Positive **C.** Zero **D.** Negative

____ **7.** An elastic collision is characterized by the conservation of:

A. Kinetic energy only
B. Linear momentum only
C. Both linear momentum and kinetic energy
D. Neither linear momentum nor kinetic energy

____ **8.** What is the cross product magnitude |A x B| if |A| = 6, |B| = 8, and the angle between them is 30 degrees?

A. 12 **B.** 24 **C.** 48 **D.** 41.6

____ **9.** The gravitational potential energy of a 5 kg object raised 10 meters above the ground is: (take g = 9.8 m/s^2)

A. 49 J **B.** 98 J **C.** 490 J **D.** 980 J

____ **10.** If vector A has magnitude 10 N directed east and vector B has magnitude 10 N directed north, what is the direction of A + B?

A. North-East (45°) **B.** North-West (135°)

C. South-East (315°) **D.** Due East (0°)

____ **11.** What is the scalar (dot) product of two non-zero vectors if they are mutually perpendicular?

A. Zero **B.** 1

C. Product of their magnitudes **D.** Negative infinity

____ **12.** A projectile is launched from ground level with speed u at angle theta. What is its total time of flight T? (g is acceleration due to gravity)

A. (2u * sin theta) / g **B.** (u * sin theta) / g

C. (u^2 * sin 2theta) / g **D.** (u * cos theta) / g

____ **13.** At the apex (highest point) of projectile trajectory, the acceleration of the object is:

A. g downwards **B.** Zero

C. g upwards **D.** Horizontal only

____ **14.** An object rotates with angular speed omega = 10 rad/s in a circle of radius r = 0.2 m. What is its linear tangential speed v?

A. 2 m/s **B.** 50 m/s **C.** 0.02 m/s **D.** 5 m/s

____ **15.** What banking angle theta of a frictionless curve of radius r prevents sideways skidding at speed v?

A. tan(theta) = v^2 / (r*g) **B.** sin(theta) = v / (r*g)

C. tan(theta) = r*g / v^2 **D.** cos(theta) = v^2 / r

____ **16.** The work-energy theorem states that the net work done on an object equals:

A. Change in kinetic energy (Delta KE) **B.** Total potential energy

C. Total momentum **D.** Rate of change of power

____ **17.** A 1000 W electric motor operates for 20 seconds. How much electrical energy does it consume?

A. 20,000 J **B.** 50 J **C.** 2,000 J **D.** 200 J

____ **18.** Linear momentum of an isolated system is conserved when:

A. The net external force acting on the system is zero

B. Internal forces are zero

C. Kinetic energy is conserved

D. All collisions are elastic

____ **19.** A 3 kg ball moving at 4 m/s collides and sticks to a stationary 1 kg ball (inelastic collision). What is their combined final velocity?

A. 3 m/s **B.** 4 m/s **C.** 2 m/s **D.** 1 m/s

____ **20.** What is the SI unit of power?

A. Watt (J/s) **B.** Joule (N*m)

C. Newton (kg*m/s^2) **D.** Pascal (N/m^2)

____ **21.** An ideal spring with spring constant k = 200 N/m is compressed by x = 0.1 m. What is the stored elastic potential energy?

A. 1.0 J **B.** 2.0 J **C.** 10 J **D.** 20 J

____ **22.** Which of the following is a non-conservative force?

A. Kinetic friction force **B.** Gravitational force

C. Electrostatic force **D.** Ideal spring force

____ **23.** The unit vector in the direction of vector V = 6i - 8j is:

A. 0.6i - 0.8j **B.** 6i - 8j **C.** 0.8i - 0.6j **D.** 10i - 10j

____ **24.** A body moving in a circle at constant speed has:

A. Constant velocity **B.** Constant speed but varying velocity

C. Zero acceleration **D.** Zero net force

____ **25.** What is the impulse delivered by a 50 N force acting on an object for 0.4 seconds?

A. 20 N*s **B.** 125 N*s **C.** 50.4 N*s **D.** 200 N*s